ANALYSIS_PLAN

ANALYSIS PLAN — EP-AGENT-001

Version: v0.2-DRAFT (2026-10-15) — DESIGN ONLY. No data exist.

1. Study character

EP-AGENT-001 is a **deterministic integration study**. For a fixed contract and fixed policy, the ExecutionProof decision is expected to be deterministic; the study asks whether four integration paths deliver that decision and its ProofRecord linkage intact. Analysis is therefore **descriptive**: exact counts, exact agreement, and distributional summaries of engineering metrics. No hypothesis tests are run on primary endpoints.

2. Primary endpoints (confirmatory)

Endpoint	Computation	Reporting
P1	Σ I(integration_complete) over 20 cells	"k/20 (x%)" plus a 4×5 cell table
P2	Σ I(observed = expected) over 20 cells	"k/20 (x%)"; every mismatch listed with framework, case, observed, expected
P3	Per case: share of frameworks agreeing with the modal verdict	P3_B1…P3_B5 each as "k/4"; any divergence listed; canonical-field and `action_hash` equality reported per case
P4	Σ I(ref returned ∧ resolves ∧ present unchanged in framework-native result) over 20 cells	"k/20"; failures listed by sub-condition
P5	Boolean	true/false with description of any core change

Decision fidelity is exact agreement; there is no tolerance. A verdict is the exact string `ALLOW`, `HOLD`, or `DENY`; anything else (including errors, timeouts, or null) is a mismatch.

Classification is applied mechanically per `PASS_FAIL_RULES.md`.

3. Secondary endpoints (descriptive)

Metric	Summary
S1 Adapter LOC	Integer per framework; shared vs framework-specific lines
S2 Setup burden	Step count; time-to-first-verify (s)
S3 Dependency burden	Added package count; installed size (MB)
S4/S5 Latency	median, p95, min, max per cell; raw values published alongside
S6 Integration friction	Count of workarounds/deviations + qualitative notes
S7 Configuration surface	Count of framework-specific config keys/objects

4. Latency protocol

- Warmup: **5** calls per case per framework, discarded.

- Sample: **30** measured calls per case per framework (600 measured calls total).
- Timer: `time.perf_counter_ns()`; boundaries defined in `STUDY_PROTOCOL.md` §7.
- Report: median and p95 (p95 = nearest-rank on the 30 samples), plus min/max and the raw vector.
- Optional: bootstrap 95% CI for the median, 1,000 resamples, percentile method, seed **20261015** (frozen v0.3; `EP_AGENT_001_SEED`). Implemented in `scripts/analysis/analyze.py`; validated only by `scripts/analysis/selftest_synthetic.py` (synthetic dummy data — not evidence).
- Label: “RF internal measurement” with host CPU, OS, Python version, and ExecutionProof deployment mode.
- Latency never affects classification.

5. Framework comparison table (structure)

Framework	Version	P1 cells	P2 cells	P4 cells	Core mod?	LOC	Steps	TTFV (s)	Added deps	S5 median (ms)	S5 p95 (ms)	Workarounds
openai_agents	0.24.0	k/5	k/5	k/5	y/n	n	n	x	n	x	x	n
langgraph	1.2.15	…	…	…	…	…	…	…	…	…	…	…
crewai	1.15.28	…	…	…	…	…	…	…	…	…	…	…
maf	1.22.0	…	…	…	…	…	…	…	…	…	…	…

Raw values are reported alongside any ranking. No framework is declared “best”; differences are reported as measured values with host context.

6. What is not done

- No inferential statistics are manufactured for trivial technical variability (e.g., no t-tests or p-values on latency differences of a few milliseconds between frameworks on one host).
- No post-hoc exclusion of cells, frameworks, or latency outliers. Outliers are reported, not removed.
- No pooled “overall score” combining primary and secondary endpoints.
- No re-analysis that changes thresholds after data are seen.

7. Deviation handling

- Deviations are logged per `STUDY_PROTOCOL.md` §10 and reported in a dedicated section of the results.
- Any analysis not specified here is labelled **exploratory** and cannot change the classification.
- If an INCONCLUSIVE condition is invoked, the analysis reports all cells that did complete, clearly marked, and names the invalidation condition with evidence.

8. Missing data

A cell with no usable decision response counts as `integration_complete = false`, a P2 mismatch, and a P4 failure, unless an INCONCLUSIVE condition is documented for that cell; in that case it is reported as invalidated, not silently dropped.